Algebra and Trigonometry **Name (Print):** ANSWER KEY
MATH 1450-01

- This exam contains 15 pages (including the front and back of this cover page) and 17 problems. Check to see if any pages are missing.
- This is the first exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 3 bonus questions worth a total of 6 possible points.

Page:	3	4	5	6	7	8	9	10	11	12	13	Total
Points:	20	10	5	10	10	10	5	5	10	10	5	100
Score:												

Bonus:	

1. Mark each statement as True or False by completely shading in the box corresponding to your answer.

For readability in the key, correct answers are colored in red, e.g. Correct Answer, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

2 pts (a) If $f(x) = (2x+2)^2$, then $f(x+h) = (2x+2)^2 + h$.

~~True~~ False

2 pts (b) The slope of the line that passes through $(6, -6)$ and $(2, -4)$ is -2

~~True~~ False

2 pts (c) The x-intercept of $y = -4x + 6$ is $\left(\frac{3}{2}, 0\right)$ and the y-intercept is $(0, 6)$.

True ~~False~~

2 pts (d) The average rate of change of $g(x) = 5x^2 - 5$ on the interval $[x, x+h]$ is $5h + 10x$.

True ~~False~~

2 pts (e) If $f(x) = \sqrt{x} + 5$ and $g(x) = x^2 + 2$, then the composition $(g \circ f)(x) = g(f(x))$ is given by $(g \circ f)(x) = \sqrt{x^2+2} + 5$.

~~True~~ False

2 pts (f) In interval notation, the domain of the function $f(x) = \frac{4x+4}{\sqrt{6-x}}$ is $(-\infty, 6)$.

True ~~False~~

2 pts (g) The function $f(x) = \frac{1}{x} + 2x$ is an even function.

~~True~~ False

2 pts (h) If $f(x)$ is a one-to-one function and $f^{-1}(-5) = -4$, then $f(-4) = -5$.

True ~~False~~

2 pts (i) The graph of $y = f(x) + 4$ is a horizontal translation of the graph of $y = f(x)$

~~True~~ False

2 pts (j) If the graph of $y = f(x)$ is the same as the graph of $y = f(-x)$, then $f(x)$ is an even function.

True ~~False~~

2. Suppose the functions p and m are given by

$$p(x) = \frac{1}{\sqrt{x}}, \text{ and } m(x) = x^2 - 4.$$

Find the domain of the functions given below. Express your answer in interval notation

$\boxed{5 \text{ pts}}$ (a) $\dfrac{p(x)}{m(x)}$

(A) $(0, 2) \cup (2, \infty)$ (B) $(-\infty, -2] \cup [2, \infty)$ (C) $(-\infty, 2) \cup (2, \infty)$ (D) $(-\infty, \infty)$

We begin by noting that the domain of $p(x)$ is $(0, \infty)$ and the domain of $m(x) = (-\infty, \infty)$. So, the values for which both $p(x)$ and $m(x)$ are defined is given by the intersection of their domains, i.e. $(0, \infty)$. To find the domain of $\dfrac{p(x)}{m(x)}$, we need only exclude from $(0, \infty)$ the values for which $m(x) = 0$. Solving, we have

$$\begin{aligned} m(x) &= 0 \\ x^2 - 4 &= 0 \\ x^2 &= 4 \\ x &= \pm 2. \end{aligned}$$

As 2 is the only one of these values which lies in $(0, \infty)$, we may conclude that the domain of $\dfrac{p(x)}{m(x)}$ is $(0, 2) \cup (2, \infty)$.

$\boxed{5 \text{ pts}}$ (b) $p\big(m(x)\big)$

(A) $(-\infty, \infty)$ (B) $(-2, 2)$ (C) $(-\infty, -2) \cup (2, \infty)$ (D) $(2, \infty)$

We begin by noting that the domain of $p(x)$ is $(0, \infty)$ and the domain of $m(x) = (-\infty, \infty)$. To find the domain of $p\big(m(x)\big)$, we need only exclude from the domain of $m(x)$ the values for which $m(x)$ does not lie in the domain of $p(x)$. That is, we solve the inequality $m(x) > 0$.

$$\begin{aligned} m(x) &> 0 \\ x^2 - 4 &> 0 \\ x^2 &> 4 \\ x < -2 &\text{ or } x > 2. \end{aligned}$$

Hence, we may conclude that the domain of $p\big(m(x)\big)$ is $(-\infty, -2) \cup (2, \infty)$.

5 pts 3. Solve the inequality

$$3|v + 2| - 4 \geq 8.$$

Express your answer in interval notation.

(A) $(-\infty, -6) \cup (2, \infty)$ (B) $(-\infty, \infty)$ (C) $[-6, 2]$ (D) $(-\infty, -6] \cup [2, \infty)$

First, we manipulate our inequality into a more convenient form:

$$\begin{aligned} 3|v + 2| - 4 &\geq 8 \\ 3|v + 2| &\geq 12 \\ |v + 2| &\geq 4. \end{aligned}$$

Recall that $|a| \geq b$ holds if either $a \leq -b$ or $a \geq b$. So, it follows that

$$|v + 2| \geq 4$$

if either

$$v + 2 \leq -4 \text{ or } v + 2 \geq 4.$$

We solve each of these inequalities individually:

$$\begin{aligned} v + 2 &\geq 4 \\ v &\geq 2 \\ v &\in [2, \infty) \end{aligned} \qquad \begin{aligned} v + 2 &\leq -4 \\ v &\leq -6 \\ v &\in (-\infty, -6] \end{aligned}$$

So, combining these solutions, it follows that $3|v+2|-4 \geq 8$ is true when v is in $(-\infty, -6] \cup [2, \infty)$.

5 pts 4. Find the value of k if the graph of a linear function $y = f(x)$ has slope $m = 3$ and passes through the points $(4, 7)$ and $(k, 1)$.

(A) $k = 2$ (B) $k = -2$ (C) $k = -6$ (D) $k = 6$

Using the point-slope form $y - y_1 = m(x - x_1)$ of a line with slope m passing through a point (x_1, y_1), we can see that our line is given by the function

$$f(x) = 3(x - 4) + 7.$$

So, since a point (a, b) is on the graph of $y = f(x)$ if and only if $f(a) = b$, we may determine the value of k by setting $f(k) = 1$ and solving for k. That is, we obtain our final answer by carrying out the following computation:

$$\begin{aligned} 3(k-4) + 7 &= 1 \\ 3(k-4) &= -6 \\ k - 4 &= -2 \\ k &= 2. \end{aligned}$$

5 pts 5. Write the equation of the line that passes through the points given below. Express your answer in slope-intercept form (i.e. in the form $y = mx + b$).

$$(r, k) \text{ and } (r + 2, k + 4)$$

(A) $y = -2x + (2r - k)$ (B) $y = 2x + (k - 2r)$ (C) $y = 2x + (2k - 2r)$ (D) $y = 2kx + kr$

To begin, we first find the slope of our line:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(k + 4) - k}{(r + 2) - r} = \frac{4}{2} = 2.$$

Now, we use the the point-slope form $y - y_1 = m(x - x_1)$ of a line with slope m passing through a point (x_1, y_1) to determine the equation of our line. We use the point (r, k) (the point $(r + 2, k + 4)$ would also work):

$$\begin{aligned} y - k &= 2(x - r) \\ y &= 2(x - r) + k \\ y &= 2x + (k - 2r) \end{aligned}$$

5 pts 6. Write an equation for the line perpendicular to $y = 2x - 1$ which passes through the point $(4, 1)$.

(A) $y = \frac{1}{2}x + 3$ (B) $y = -2x + \frac{1}{3}$ (C) $y = -\frac{1}{2}x + \frac{3}{2}$ (D) $y = -\frac{1}{2}x + 3$

Recall that a line $y = m_1x + b_1$ is perpendicular to a line $y = m_2x + b_2$ if $m_1m_2 = -1$. So, since we are seeking a line which is perpendicular to $y = 2x - 1$, the slope of this line, denoted by m, must satisfy $2m = -1$. Solving, we find that $m = -\frac{1}{2}$. Now, we use the the point-slope form $y - y_1 = m(x - x_1)$ of a line with slope m passing through a point (x_1, y_1) to determine the equation of our line:

$$\begin{aligned} y - 1 &= -\frac{1}{2}(x - 4) \\ y &= -\frac{1}{2}(x - 4) + 1 \\ y &= -\frac{1}{2}x + 3. \end{aligned}$$

5 pts 7. Suppose a town has a population of $1,000$ in the year 1999, and suppose the population of the town grows at a constant rate of 125 per year. Find the equation of the line that models the town's population P as a function of t, where t is the number of years since the model began.

(A) $P = 125(t - 1999) + 1000$ (B) $P = -125t + 1000$ (C) $P = 125t + 1250.$ (D) $P = 125t + 1000$

Since we are modeling the population of the town with a linear model, the linear function we will produce will be of the form $P = mt + b$ for some real numbers m and b.

Since our model is linear, the rate of change of the population will be the slope of the line used in our model. That is, since the population of the town grows at a constant rate of 125 per year, $m = 125$.

Since the variable t corresponds to the number of years since the model began, it must be the case that $1,000 = m(0) + b = b$. So, b must be the initial population of the town, i.e. $b = 1,000$.

Combining these observations, we reach our final answer and may conclude that our linear model is given by:

$$P = 125t + 1,000.$$

8. Perform the following operations and express the results as a simplified complex number (i.e. your answer should be a complex number of the form $a + bi$ where a and b are real numbers):

$\boxed{5 \text{ pts}}$ (a) $(4 + i)(2i)$

(A) $2 - 8i$ (B) $8 - 2i$ **(C)** $-2 + 8i$ (D) $2 + 8i$

$$\begin{aligned}(4+i)(2i) &= (4)(2i) + (i)(2i)\\ &= 8i + 2i^2\\ &= 8i - 2\\ &= -2 + 8i\end{aligned}$$

$\boxed{5 \text{ pts}}$ (b) $\dfrac{4 - 2i}{2 + 2i}$

(A) $\dfrac{1}{2} - \dfrac{3}{2}i$ (B) $2 - \dfrac{3}{2}i$ (C) $\dfrac{1}{2} + \dfrac{2}{3}i$ (D) $\dfrac{3}{2} - \dfrac{1}{2}i$

$$\begin{aligned}\frac{4-2i}{2+2i} &= \left(\frac{4-2i}{2+2i}\right)\left(\frac{2-2i}{2-2i}\right)\\ &= \frac{(4-2i)(2-2i)}{2^2+2^2}\\ &= \frac{1}{8}\big((4-2i)(2-2i)\big)\\ &= \frac{1}{8}\big(8 - 4i - 8i + 4i^2\big)\\ &= \frac{1}{8}\big(8 - 12i - 4\big)\\ &= \frac{1}{2} - \frac{3}{2}i\end{aligned}$$

5 pts 9. Solve the following equation over the complex numbers:

$$x^2 - 2x + 10 = 0.$$

(A) $x = 3 \pm i$ (B) $x = 1 \pm 3i$ (C) $x = \frac{1}{3} \pm 3i$ (D) $x = 1 \pm \frac{1}{3}i$

We use the quadratic formula to solve:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(10)}}{2(1)} \\ &= \frac{2 \pm \sqrt{-36}}{2} \\ &= \frac{2 \pm 6i}{2} \\ &= 1 \pm 3i. \end{aligned}$$

5 pts 10. Use the information given below about the graph of a polynomial function $f(x)$ to determine the function. Assume the leading coefficient is 1 or -1.
HINT: Use the vertex form of a quadratic function $y = a(x-h)^2 + k$

(1) The y-intercept is $(0, 49)$
(2) The x-intercepts are $(-7, 0)$ and $(7, 0)$
(3) As $x \to -\infty$, $f(x) \to -\infty$
(4) As $x \to \infty$, $f(x) \to -\infty$
(5) The degree of $f(x)$ is 2

(A) $f(x) = x^2 - 49$ (B) $f(x) = -x^2 + 49$ (C) $f(x) = -(x-7)(x+7)$ (D) $f(x) = (x-7)(x+7)$

Since $f(x)$ must be a polynomial of degree two by requirement (5), it follows that $f(x)$ must be of the form

$$f(x) = a(x-h)^2 + k,$$

where a is the leading coefficient of f and (h, k) is the vertex of the graph of $y = f(x)$.
a
Since $f(x)$ must have a leading coefficient of 1 or -1, note that requirements (3) and (4) tell us that $a = -1$, since the graph of $y = f(x)$ must be a parabola opening downward. So,

$$f(x) = -(x-h)^2 + k.$$

Now, we need only determine the vertex (h, k) of the graph of $y = f(x)$. Requirement (2) gives us h, as the vertex must lie in the middle of the x-intercepts. That is, $h = \frac{-7+7}{2} = 0$ and

$$f(x) = -x^2 + k.$$

We can use requirement (1) to find k, since (1) tells us that f must satisfy $f(0) = 49$. Solving, we have

$$\begin{aligned} f(0) &= 49 \\ -(0)^2 + k &= 49 \\ k &= 49. \end{aligned}$$

So, we have our answer: $f(x) = -x^2 + 49$. Equivalently, answer (C) is also correct, since

$$-(x-7)(x+7) = -(x^2 - 49) = -x^2 + 49.$$

5 pts 11. Find the zeros of the following function $f(x)$ and give the multiplicity of each:

$$f(x) = (x^3 - 5x^2 + 6x)(12x^3 - 4x^4).$$

To find the zeros of $f(x)$, we first factor $f(x)$ as follows:

$$\begin{aligned} f(x) &= (x^3 - 5x^2 + 6x)(12x^3 - 4x^4) \\ &= x(x^2 - 5x + 6)(12x^3 - 4x^4) \\ &= x(x^2 - 5x + 6)(-4x^3)(x - 3) \\ &= -4x^4(x^2 - 5x + 6)(x - 3) \\ &= -4x^4(x - 2)(x - 3)(x - 3) \\ &= -4x^4(x - 2)(x - 3)^2. \end{aligned}$$

With this, we may conclude that $f(x)$ has a zero at $x = 0$ of multiplicity 4, a zero at $x = 2$ of multiplicity 1, and a zero at $x = 3$ of multiplicity 2.

5 pts 12. Determine the end behavior of the function $f(x)$ given by

$$f(x) = -4x^4 - 3x^2 + 3x - 2.$$

Since $f(x)$ is a polynomial function of even degree and has a negative leading coefficient, we may conclude that as $x \to -\infty$, $f(x) \to -\infty$ and as $x \to \infty$, $f(x) \to -\infty$.

13. Suppose $f(x)$ and $g(x)$ are the polynomial functions given by

$$f(x) = x^3 + 4x^2 + 5x + 2, \text{ and } d(x) = x + 2.$$

$\boxed{\text{5 pts}}$ (a) Find polynomials $q(x)$ and $r(x)$ such that $f(x) = d(x)q(x) + r(x)$, where $r(x)$ is either 0 or a polynomial of degree strictly less than the degree of $d(x)$.

A $q(x) = x^2 + 2x + 1$**,** $r(x) = 0$

B $q(x) = x^3 + 3x^2 + 5x + 6$, $r(x) = x - 2$

C $q(x) = x^3 + 3x^2 + 4x + 4$, $r(x) = x - 1$

D $q(x) = x^3 + 3x^2 + x - 2$, $r(x) = x + 2$

Using either polynomial long division or synthetic division, we find that $q(x) = x^2 + 2x + 1$ and $r(x) = 0$. That is, $x^3 + 4x^2 + 5x + 2$ has $x + 2$ as a factor.

$\boxed{\text{5 pts}}$ (b) Using the Remainder Theorem and your answer from part (a), compute $f(-2)$ **without** explicitly evaluating $f(-2)$ (that is, *without* plugging in -2 for x in $x^3 + 4x^2 + 5x + 2$ and solving for a value).

A $f(-2) = -2$

B $f(-2) = 2$

C $f(-2) = 0$

D $f(-2) = 1$

By the Remainder Theorem, if $f(x) = (x - k)q(x) + r(x)$, then $f(k) = r(x)$ (that is, $r(x)$ is a constant function equal to $f(k)$). So, we may conclude that $f(-2) = r(x) = 0$.

5 pts 14. Suppose $f(x)$ is the polynomial function

$$f(x) = x^3 - x^2 - 26x + 20.$$

Given that $(x+5)$ is a factor of $f(x)$, use the factor theorem to determine all of the real zeros of $f(x)$.

(A) $x = -5$, $x = 3 + \sqrt{5}$, $x = 3 - \sqrt{5}$

(B) $x = -5$, $x = 3 + \sqrt{5}$, $x = -3 + \sqrt{5}$

(C) $x = 5$, $x = \sqrt{5} + 3$, $x = \sqrt{5} - 3$

(D) $x = \sqrt{5}$, $x = 3$, $x = -3$

To find the zeros of $f(x)$ via the Factor Theorem, we begin by using either long division or synthetic division to divide the polynomial $x^3 - x^2 - 26x + 20$ by $x + 5$. Doing so, we find that $x + 5$ divides $x^3 - x^2 - 26x + 20$ evenly (as we should expect, since $x + 5$ is a factor of $f(x)$). That is, using the Division Algorithm to write the polynomial as the product of the divisor and the quotient, we have:

$$x^3 - x^2 - 26x + 20 = (x+5)(x^2 - 6x + 4).$$

Now, we use the quadratic formula on $x^2 - 6x + 4$ to get the final two zeros:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(4)}}{2(1)} \\ &= \frac{6 \pm \sqrt{20}}{2} \\ &= 3 \pm \sqrt{5}. \end{aligned}$$

By the Factor Theorem, the zeros of $f(x)$ are $x = -5$, $x = 3 + \sqrt{5}$, and $x = 3 - \sqrt{5}$.

2 (bonus) 15. Suppose the rational function $f(x)$ is given by $f(x) = \dfrac{5x+7}{2x+4}$. Find $f^{-1}(x)$.

Writing $y = f(x)$, we have $y = \dfrac{5x+7}{2x+4}$. To find $f^{-1}(x)$, we interchange x and y and solve for y:

$$\begin{aligned} x &= \frac{5y+7}{2y+4} \\ x(2y+4) &= 5y+7 \\ 2xy+4x &= 5y+7 \\ 2xy-5y &= 7-4x \\ (2x-5)y &= 7-4x \\ y &= \frac{7-4x}{2x-5}. \end{aligned}$$

So, replacing y with $f^{-1}(x)$, we have found our inverse function: $f^{-1}(x) = \dfrac{7-4x}{2x-5}$.

2 (bonus) 16. Suppose $f(x)$ is the rational function given by

$$f(x) = \frac{x^2+3x+2}{x^2-5x+6}.$$

Find the vertical asymptotes of $f(x)$.

We begin by factoring the numerator and denominator of $f(x)$ as follows:

$$\begin{aligned} f(x) &= \frac{x^2+3x+2}{x^2-5x+6} \\ &= \frac{(x+1)(x+2)}{(x-2)(x-3)}. \end{aligned}$$

The zeros of the denominator are $x = 2$ and $x = 3$. Since neither of these values are also zeros of the numerator, we may conclude that $f(x)$ has vertical asymptotes of $x = 2$ and $x = 3$.

2 (bonus) 17. Find the inverse of the function $f(x) = x^2 + 4x + 3$ over the domain $[-2, \infty)$. The domain of the inverse function will be $[-1, \infty)$.

First, we complete the square and factor $f(x)$:

$$\begin{aligned} f(x) &= x^2 + 4x + 3 \\ &= x^2 + 4x + 4 - 1 \\ &= (x+2)^2 - 1. \end{aligned}$$

Replace x by y, replace $f(x)$ by x, and solve for x (or, replace $f(x)$ with y and solve for x, if you prefer):

$$\begin{aligned} x &= (y+2)^2 - 1 \\ x + 1 &= (y+2)^2 \\ \sqrt{x+1} &= y + 2 \\ \sqrt{x+1} - 2 &= y. \end{aligned}$$

So, renaming y as $f^{-1}(x)$, we have found our inverse function:

$$f^{-1}(x) = \sqrt{x+1} - 2.$$